

SERVER TOOLKIT • UNJS • v3 BETA

Nitro v3

Write your server once.
Let the build target the platform.

Mateus • 5-minute talk • live demo included

WHAT IT IS

A build tool for servers

It does not want to own your request handler. It wants to own your deploy.

- ▶ **Web standards, not abstractions** — Request, Response, Headers, URL. h3 v2 was rewritten around them.
- ▶ **Filesystem routing** — compiled at build time, so there is no runtime router to pay for.
- ▶ **Runtime-agnostic** — one codebase for Node, Bun, Deno, Workers, Lambda.
- ▶ **Bring your own framework** — Hono or Elysia as the entry, Nitro still does the build.

Nuxt 5

ships Nitro v3 at its core

TanStack Start

built on Nitro

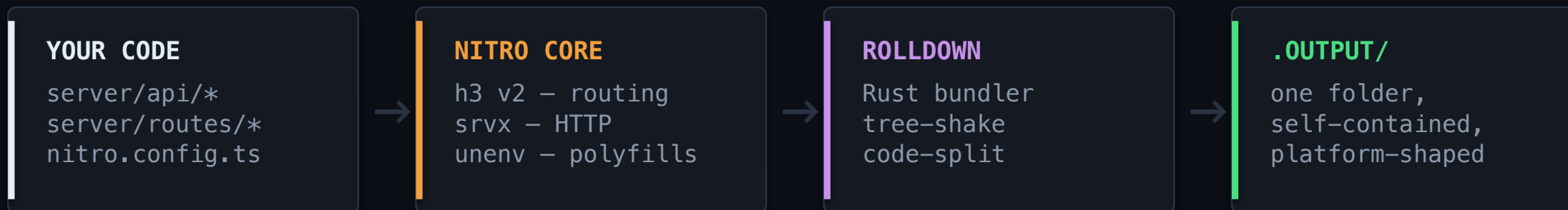
Analog · SolidStart

built on Nitro

unjs ecosystem: h3 · srvx · unstorage · ocache · db0 · unenv · rolldown

HOW IT WORKS

It compiles your server



Batteries are opt-in and tree-shaken: KV storage, SWR caching, SQL, WebSockets, scheduled tasks.

Don't import it and it is not in your bundle.

25

transitive deps for nitro v3 (measured)

273

transitive deps for nitropack v2 (measured)

22 ms

dev rebuild in my POC

One codebase, five builds

Measured on the POC in this repo. Same source files, nothing changed between runs.

```
NITRO_PRESET=cloudflare_module npx nitro build
```

preset	output size	gzipped	runs on
node (default)	114 kB	33.3 kB	✓ verified locally
cloudflare_module	88.2 kB	26.6 kB	✓ build passes
vercel	87.5 kB	26.3 kB	✓ build passes
deno_deploy	95.7 kB	29.0 kB	✓ build passes
bun	96.2 kB	29.3 kB	✓ build passes

Routing is the filename. Caching is a wrapper.

filesystem routing

```
// server/api/hello.get.ts -> GET /api/hello
import { defineHandler } from "nitro";

export default defineHandler((event) => {
  // event.req is a real web Request
  return { hello: "Nitro",
    method: event.req.method };
});

// hello.post.ts -> POST, same path.
// No router file. Nothing to keep in sync.
```

stale-while-revalidate in 10 lines

```
// server/api/stars/[...repo].ts
const cachedStars = defineCachedFunction(
  async (repo: string) => {
    const r = await fetch(gh + repo);
    return (await r.json()).stargazers_count;
  },
  { name: "ghStars",
    maxAge: 60, // fresh for 60s
    staleMaxAge: -1, // then stale + bg
    getKey: (repo) => repo },
);

// No redis client. No conn string.
```

[...repo] is a catch-all param · /api/stars/nitrojs/nitro -> repo = "nitrojs/nitro"

Live: the whole server is seven files

1. **stars/[...repo].ts** no redis client → ~300 ms becomes 0.4 ms
2. **nitro.config.ts** cors, cache, proxy, auth — as config
3. **routes/admin/** password on the route, no auth code in the file
4. **nitro build ×5** node · cloudflare · vercel · deno · bun

localhost:3100 live code, live cache, live builds

What it costs you

WORTH IT

- + No lock-in — platform is a build flag
- + Cache, KV, SQL, WebSockets included
- + Tiny dep tree: 25 vs 273 packages
- + Sub-30 ms dev rebuilds (Rolldown)
- + Web standards — knowledge transfers out

WHAT BIT ME

- v3 is beta — API moves, docs lag
- No jiti installed → config won't load
- serverDir defaults to false → silent 404s
- Portable code ≠ portable dependencies
- Smaller community than Express

Every item on the right is something I hit building the POC for this talk.

Where Nitro actually sits

	deps*	multi-runtime	deploy presets	cache + KV built in
nitro v3	25	✓ Node/Bun/Deno/edge	✓ ~20	✓ yes
hono	1	✓ Node/Bun/Deno/edge	✗ none	✗ bring your own
fastify	41	✗ Node-centric	✗ none	✗ plugins
express	65	✗ Node only	✗ none	✗ bring your own
next.js	~30	~ Node + edge	~ Vercel-shaped	✓ yes

Not either-or: Nitro will run Hono, Elysia or plain h3 as its server entry — you keep the router you like and still get the presets.

** transitive package count, measured with a clean npm install on 2026-07-29.*

VERDICT

Deployment as a compile target

That is the idea worth stealing, whether or not you adopt the tool.

REACH FOR IT WHEN

- Edge or multi-platform is on the table
- You want cache + KV without provisioning
- You're already in the Nuxt / unjs world

SKIP IT WHEN

- Production ships next week (wait for stable)
- A plain Node service on one host is enough
- Your team needs a big hiring pool / lots of docs

nitro.build

github.com/nitrojs/nitro

Demo source, measured numbers and the runbook are in this repo — ask me for it.

Questions?